

Mental Health Analysis

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OUTLINE

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Abstract

Mental health analysis is a crucial step in understanding the factors that contribute to an individual's well-being. This study delves into various lifestyle, demographic, and health-related factors to identify patterns that may influence mental health conditions, particularly depression. By analyzing key indicators such as sleep patterns, dietary habits, physical activity, and medical history, we aim to uncover correlations that could help in early detection and prevention strategies. The findings from this analysis can support healthcare professionals, policymakers, and researchers in developing more effective interventions and awareness programs to promote mental well-being in society.

Problem Statement

Mental health is a huge concerns of current era and depression is a leading cause of disability worldwide, affecting millions of individuals across various demographics. Despite increased awareness, early detection and intervention remain significant challenges. The goal of this analysis is to examine a depression dataset to identify key factors influencing mental health conditions and detect patterns , which will help in better understanding of these domain.

Aim and Objective

AIM:

To analyze depression data and identify key factors influencing mental health conditions, enabling better understanding of factors affecting mental health

OBJECTIVE:

- Data Exploration & Understanding:
- Trend Analysis & Visualization
 - Examine demographic and behavioral factors affecting depression.
 - Identify correlations between variables such as age,employment,dietary habits, and mental health conditions.
- Provide actionable insights to improve mental health awareness and support.

Proposed Solution

1. Data Collection

- Gather a reliable depression dataset.
- Reduced the dataset for convenience.

2. Preprocessing and cleaning

- Clean the data by handling missing values, outliers, and inconsistencies.
- Standardize and normalize relevant features for accurate analysis.

3. Exploratory Data Analysis (EDA)

- Identify key demographic, behavioral, and social factors linked to depression.
- Use statistical analysis and visualizations (e.g., histograms, correlation matrices, heatmaps) to detect trends and patterns.
- Compare depression prevalence across different population groups.

4. Feature Engineering & Selection

- Identify the most influential variables contributing to depression.
- Create new derived features if necessary (e.g., stress levels, sleep patterns, lifestyle habits).
- Use feature selection techniques to retain the most important predictors.

5. Summarize the data insights

- Summarize all the factors of people with depression so that it can provide actionable insights for healthcare professionals, policymakers, and communities.

System Architecture

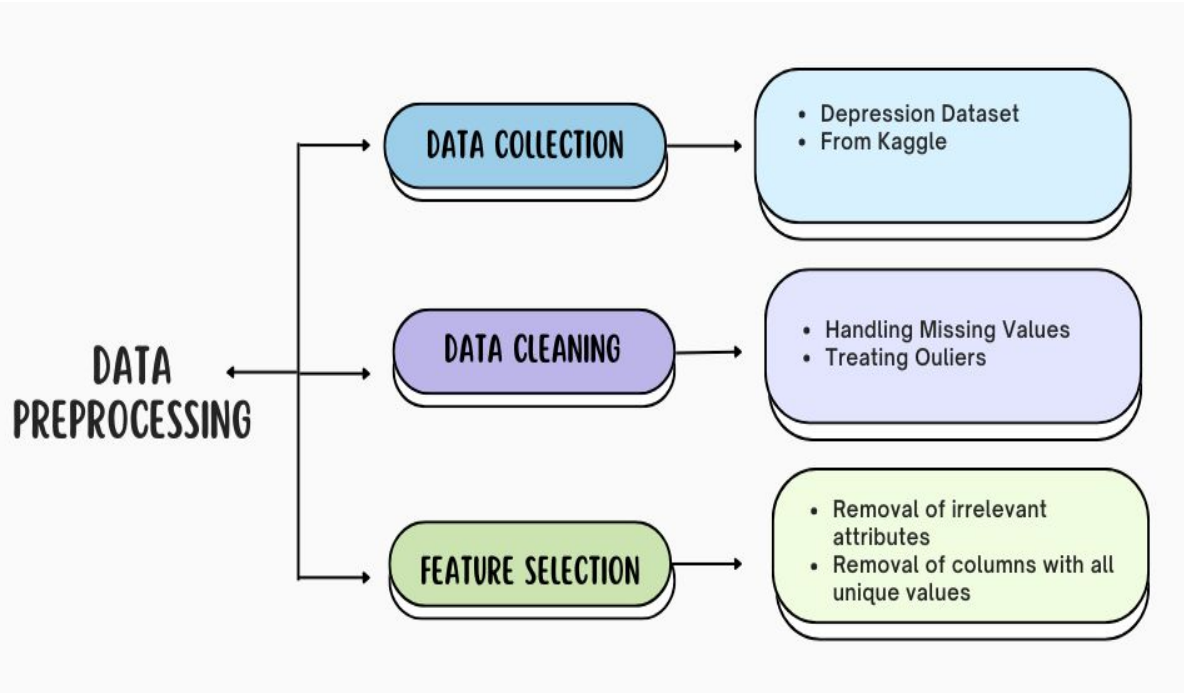
1. Data Ingestion Layer

- Data Sources : Depression Dataset (Kaggle, potentially reduced by ChatGPT)
- Data Type : Comma Separated values (csv)
- Data Structured : Dataframe (by Pandas Library)
- Data Read : Use `pandas.read_csv()` in Python to load the dataset.

2. Data Preprocessing Layer

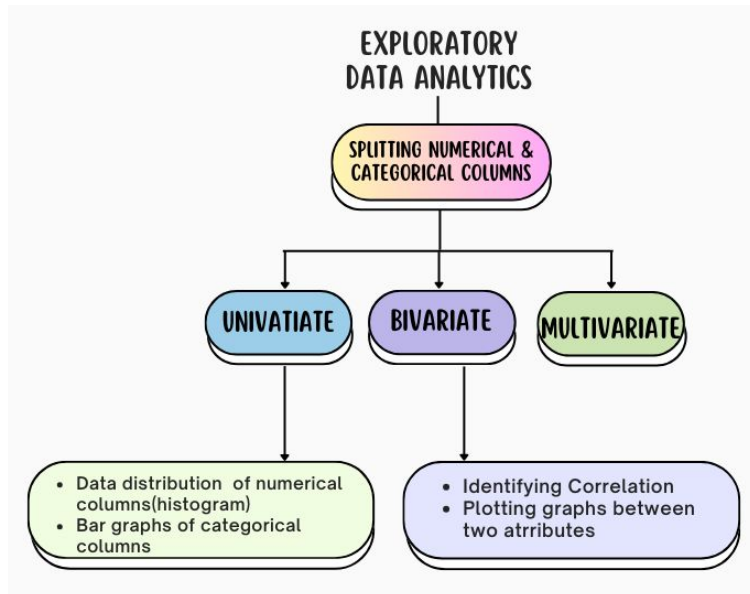
- Data Understanding:
 - Use `.head()`, `.tail()`, and `.shape()` to understand the dataset structure.
 - `info()` and `describe()` to understand data

- Data Cleaning:
 - Handle missing values (imputation or removal).
 - Address outliers (removal or transformation).
 - Correct inconsistencies in data.



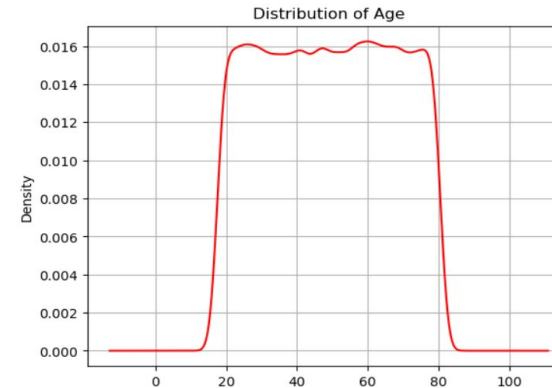
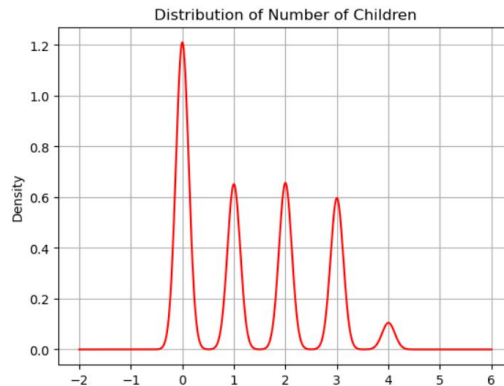
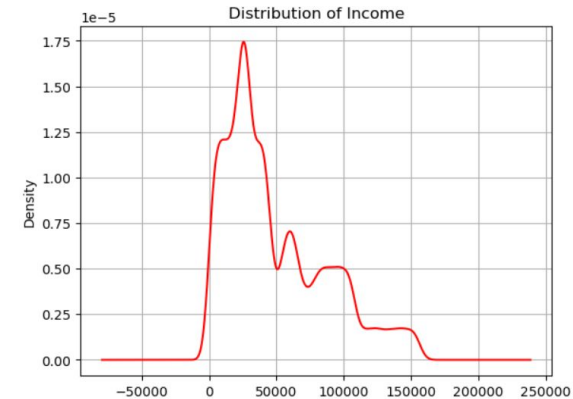
3. Exploratory Data Analysis (EDA) Layer

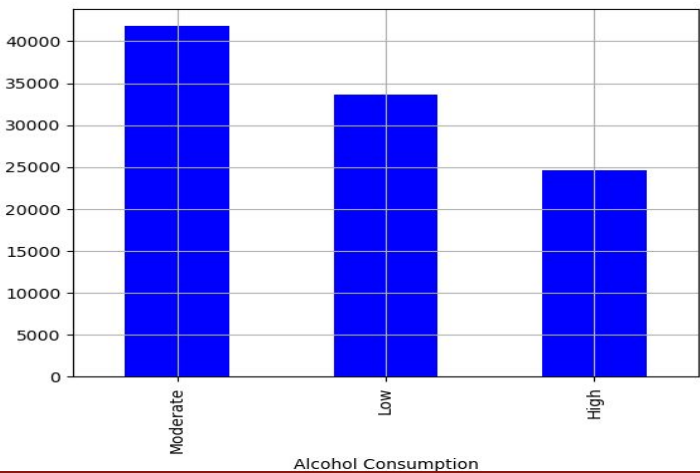
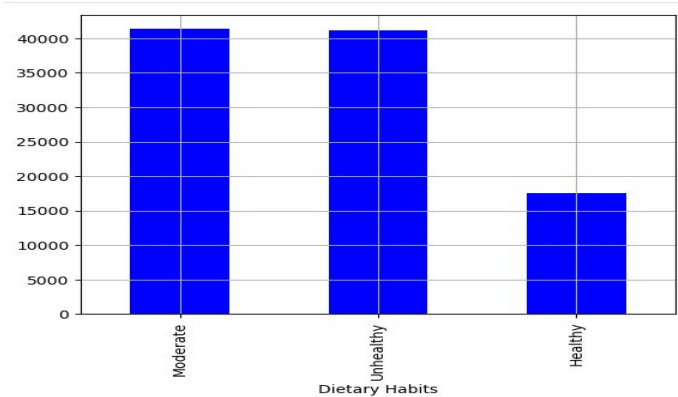
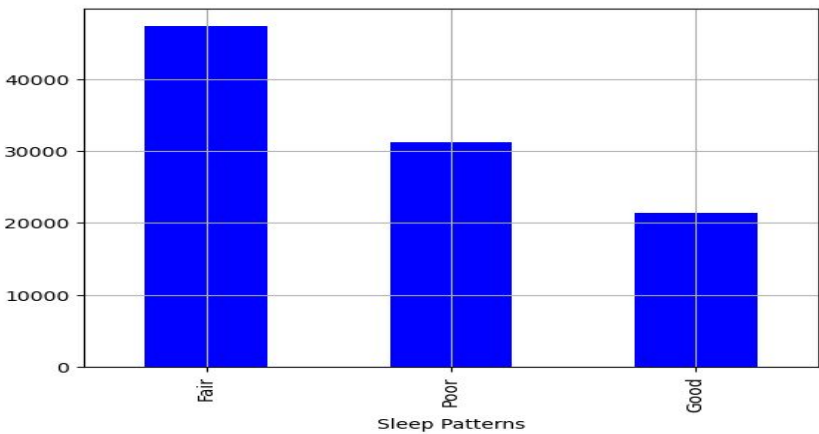
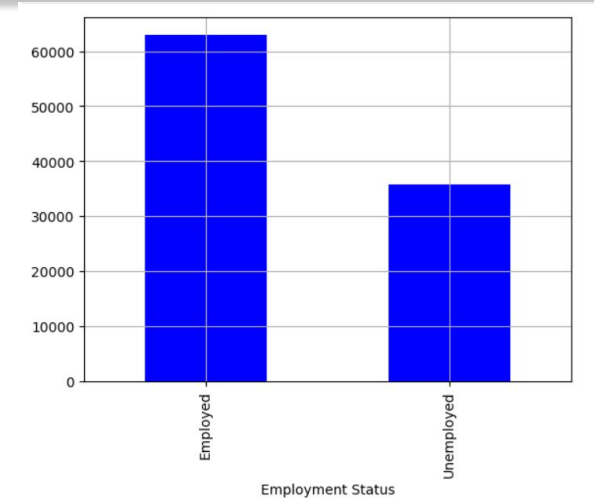
- Calculate descriptive statistics (mean, median, standard deviation, etc.).
- Perform univariate and bivariate analysis
- Create histograms to visualize data distributions.
- Use other visualizations to (e.g., histograms, correlation matrices, heatmaps) identify trends and patterns.

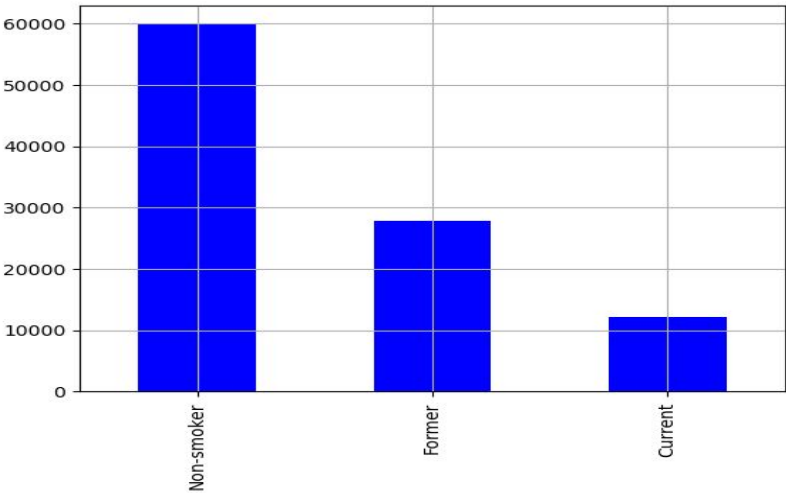


System Deployment Approach

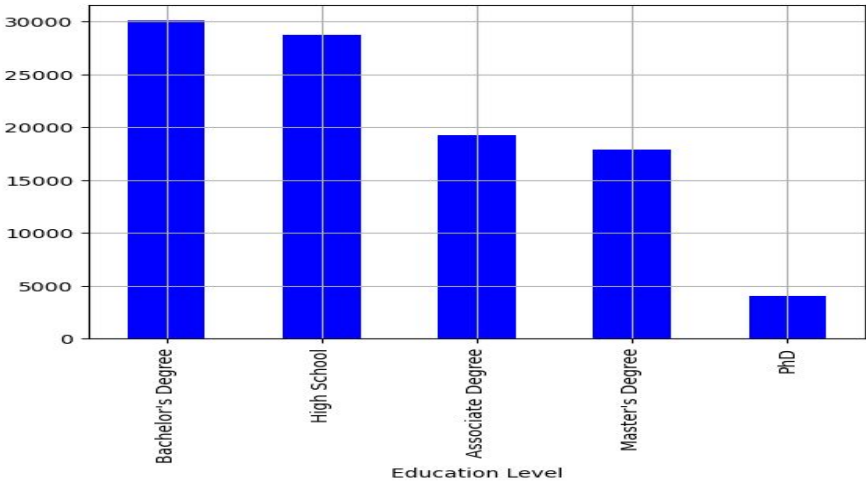
- Data Ingestion
- Data Preprocessing : No missing values found
 'Income' outliers have outliers
 Eliminate 'Name' attribute
- Exploratory Data Analysis (EDA) : Statistical analysis
 Visualization



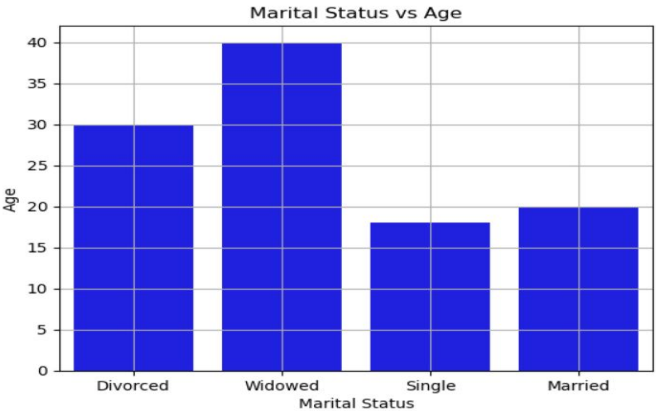




Smoking Status



Education Level



Marital Status

Insights :

- Possible age range to acquire depression is 18 to 80
- Married person have highest the highest to acquire depression and divorced person have lower possibility
- People with high school education and Bachelor degree are highest number of people with depression
- Majority of people are non smokers
- People with sedentary physical activity are more prone to be depress
- Employed are more depressed than unemployed person
- People with high alcohol consumption are the least depressed and moderate consumers are the most depressed
- People with healthy diet have the least possibility to get depressed while people with unhealthy and moderate diet have highest possibility
- People with Good sleep pattern have the least possibility to get depressed
- People with no history of mental illness are the most depressed
- Elderly widows are more depressed

Conclusion

The results of this project assist medical professionals, researchers, and policymakers in making decisions about mental health awareness and treatment strategies. Moreover, it also helps professionals to bring AI/ML based solutions which leads to early intervention and detection strategies in the mental health domain.

Future Scope

Although the present project sets a solid ground for depression analysis, there remains vast scope for additional enhancements and extensions, including:

- **Integration with Real-World Data Sources:** Subsequent research can integrate real-time data from mental health mobile applications, surveys, or electronic medical records.
- **AI/ML Implementation** - Analysis can be used for ML model fitting or AI implementation.
- **Incorporating Deep Learning:** State-of-the-art deep learning architectures like transformers and LSTMs may be investigated to make more precise predictions.

- **Personalized Mental Health Recommendations:** AI-based personalized treatment could be established to offer personalized mental health interventions.
- **Cross-Cultural Analysis:** Increasing the size of the dataset to encompass multicultural populations may add to the results' generalizability.

Reference

- Title : Prediction of Mental Health Based on Data Science,PaperId: 20914,Published in: International Journal Of Advance Research And Innovative Ideas In Education,Publisher: IJARIIIE,e-ISSN: 2395-4396,Volume/Issue: Volume 9 Issue 3 2023, DOI: 16.0415/IJARIIIE-20914, Licence: : IJARIIIE is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
- Mental Health Prediction using Sentiment Analysis,Authors: Rishabh Verma, Nipun , Nitin Rana, Dr. Rakesh Kumar Arora, DOI Link: <https://doi.org/10.22214/ijraset.2023.57534>
- Statistical procedures for analyzing mental health services data, Authors - Jon D. Elhai a, Patrick S. Calhoun b, Julian D. Ford c, DOI Link : <https://doi.org/10.1016/j.psychres.2007.07.003>
- Dataset : Depression Dataset([Depression Dataset](#)) From kaggle by Anthony Therrien , further reduced by Chat GPT

Thank you!